

Topological Collapse and Asymmetric Divergence of Carmichael Pseudoprimes on the $6N$ Skeleton*

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Abstract

On the $6N$ skeleton every integer coprime to 6 sits on a wing: the right wing $6N+1 \pmod{6}$ or the left wing $6N-1 \pmod{6}$. We read the Carmichael numbers through this geometry. Korselt's criterion forces a sharp, elementary structure that we state and verify: (i) a one-way semipermeable membrane—every Carmichael number on the left wing has *all* of its prime factors on the left wing (and is not divisible by 3); and (ii) a parity law—a Carmichael number coprime to 6 lies on the left wing iff it has an odd number of left-wing prime factors. Enumerating all 1547 Carmichael numbers up to 10^{10} , we find these two laws hold with *zero* violations. The geometric consequence is a violent asymmetry: the right-to-left ratio $R(X) = \#\{\text{right}\}/\#\{\text{left}\}$ climbs monotonically from 5.0 at $X = 10^4$ to 88.8 at $X = 10^{10}$, a collapse of the left-wing population across seven decades. We trace the mechanism to a single structural fact: the left wing admits only *monochromatic* products (all factors $\equiv 5 \pmod{6}$) of *odd* length, while the mean number of prime factors $\bar{k}(X)$ of Carmichael numbers grows ($3.00 \rightarrow 4.23$ over the range), so the monochromatic configuration becomes exponentially scarce. We are explicit about the limit of the claim: the *rate* of divergence is *not* identifiable. Three functional forms—a power of $\log X$, a small power of X , and a stretched exponential—fit the seven points equally well, and the left-wing counts are single- to low-double-digit, so Poisson noise and the deep pre-asymptotic character of Carmichael statistics make the asymptotic law unrecoverable in any humanly reachable computational window. The structure (membrane, parity) and the phenomenon (asymmetric collapse) are firm; the asymptotic divergence law is, by construction, beyond reach.

1 Introduction

A Carmichael number is a composite n for which $a^n \equiv a \pmod{n}$ for every integer a ; by Korselt's criterion [1] these are exactly the squarefree n with at least three prime factors such that $p-1 \mid n-1$ for every prime $p \mid n$. They are infinite in number [3], but their finer distribution—the growth of $C(x) = \#\{\text{Carmichael} \leq x\}$ and the internal correlations among their prime factors—remains, in Erdős's heuristic [4] and in Pinch's tabulations [5], only partly understood.

The present series reads arithmetic objects through the $6N$ skeleton, on which every prime > 3 lies on a fault line $6N \pm 1$. Here we ask what Korselt's criterion looks like in that geometry. The answer is an elementary but unusually rigid structure, and a measured macroscopic consequence. We make no claim that the structural facts are new as mathematics—they follow in a few lines

*Code and measured data: <https://github.com/Ruqing1963/6N-carmichael-collapse> (Part XXXVI).

from Korselt—only that their statement in the wing coordinates is clean, and that the resulting population asymmetry, and its honest non-identifiability, are worth recording.

2 The two structural laws

Write a prime $p > 3$ as *right* if $p \equiv 1 \pmod{6}$ and *left* if $p \equiv 5 \pmod{6}$. On the units $(\mathbb{Z}/6)^\times = \{1, 5\} \cong \{\pm 1\}$, right primes carry the residue $+1$ and left primes the residue -1 .

Proposition 1 (The semipermeable membrane). *Let n be a Carmichael number with $n \equiv 5 \pmod{6}$ (the left wing). Then $3 \nmid n$ and every prime factor of n satisfies $p \equiv 5 \pmod{6}$. No right-wing prime, and no factor of 3, can occur.*

Proof. $n \equiv 5 \pmod{6}$ gives n odd and $3 \nmid n$, so n is coprime to 6 and every prime factor is $\equiv 1$ or $\equiv 5 \pmod{6}$. Suppose $p \mid n$ with $p \equiv 1 \pmod{6}$. Then $6 \mid p - 1$. By Korselt $p - 1 \mid n - 1$, hence $6 \mid n - 1$. But $n \equiv 5 \pmod{6}$ gives $n - 1 \equiv 4 \pmod{6}$, so $6 \nmid n - 1$ —a contradiction. Thus no factor is $\equiv 1 \pmod{6}$, and all factors are $\equiv 5 \pmod{6}$. $\square$

Proposition 2 (The parity law). *Let n be a Carmichael number coprime to 6, with ℓ prime factors $\equiv 5 \pmod{6}$ (counted with the multiplicity 1, as n is squarefree). Then $n \equiv (-1)^\ell \pmod{6}$. Equivalently: n is on the right wing iff ℓ is even, and on the left wing iff ℓ is odd.*

Proof. Modulo 6 the residue of a product of units is the product of residues; right factors contribute $+1$ and left factors -1 , so $n \equiv (+1)^{\#\text{right}}(-1)^\ell \equiv (-1)^\ell \pmod{6}$. $\square$

Corollary 1. *A left-wing Carmichael number is a product of an odd number (≥ 3) of distinct primes, all $\equiv 5 \pmod{6}$. The left wing is the single monochromatic, odd-length family; the right wing collects everything else—the all-right products, and all mixed products with an even number of left factors.*

These are the entire structural content, and they are elementary. Their force is geometric: the left wing is closed under Korselt to a single, narrow family, while the right wing is an open basin.

3 The measurement

We enumerated *all* Carmichael numbers up to 10^{10} by the Korselt divisor method (for each ordered prefix of prime factors with product M , the final prime r must satisfy $(r - 1) \mid (M - 1)$, checked against $p - 1 \mid n - 1$ for the remaining factors), recovering the known count $C(10^{10}) = 1547$ [5] exactly, with the correct factorisation of each. Across all 1547 numbers, Propositions 1 and 2 hold with **zero violations**: every left-wing Carmichael number is monochromatically left, and every right-wing Carmichael number has an even number of left-wing factors.

Table 1 reports the cumulative wing census by decade. The right-to-left ratio $R(X) = \#\{n \leq X : n \equiv 1\} / \#\{n \leq X : n \equiv 5\}$ rises monotonically from 5.0 to 88.8. (A third, off-skeleton class with $3 \mid n$, e.g. $561 = 3 \cdot 11 \cdot 17$ at residue 3 mod 6, is small and is excluded from the wing ratio.)

4 Mechanism: monochromatic suppression under a growing factor count

By Corollary 1 a left-wing Carmichael number must be *monochromatic*: all of its k prime factors must fall on the left wing, with k odd. For a Carmichael number with k factors, the heuristic

$\log_{10} X$	$C(X)$	right	left	$R(X)$	$\bar{k}(X)$
4	7	5	1	5.00	3.00
5	16	13	1	13.00	3.25
6	43	38	2	19.00	3.49
7	105	98	4	24.50	3.58
8	255	243	6	40.50	3.78
9	646	623	12	51.92	4.00
10	1547	1510	17	88.82	4.23

Table 1: Cumulative wing census of Carmichael numbers. $R(X)$ is the right/left ratio; $\bar{k}(X)$ is the mean number of prime factors. The left-wing population collapses relative to the right as X grows.

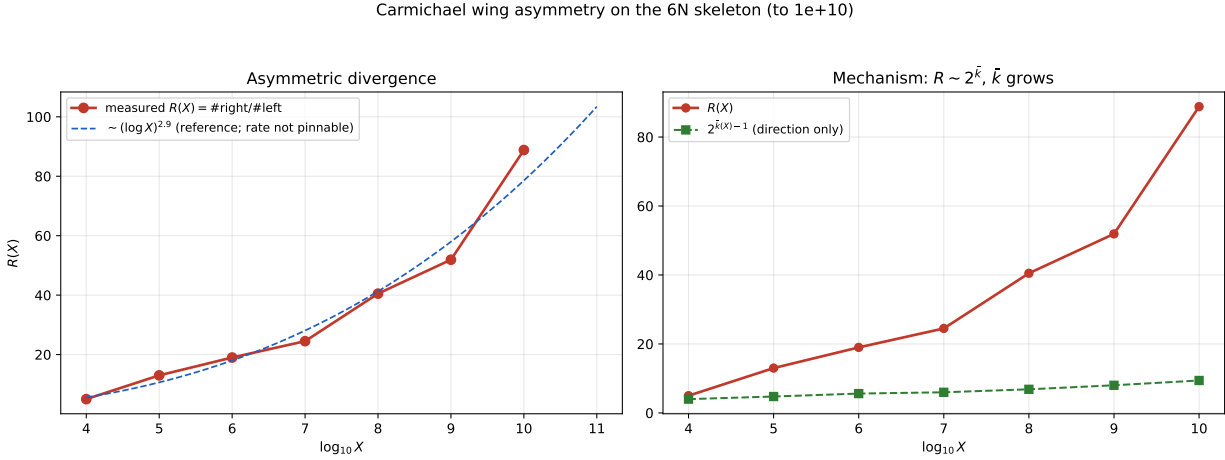


Figure 1: (Left) The measured asymmetric divergence $R(X)$ over seven decades, with a best-fitting power of $\log X$ drawn for reference only—the rate is not identifiable (§5). (Right) The mechanism: $R(X)$ tracks the growth of the mean prime-factor count $\bar{k}(X)$ through the monochromatic suppression $\sim 2^{-\bar{k}}$; the curve $2^{\bar{k}-1}$ captures the direction but undershoots the measured magnitude, because the right wing additionally absorbs the pure-right and even-mixed families.

chance that all k are left-wing is of order 2^{-k} (the two wings carry primes of nearly equal density). The mean number of prime factors of Carmichael numbers grows with scale—we measure $\bar{k}(X)$ rising from 3.00 to 4.23 across the range (Table 1)—so the monochromatic condition becomes exponentially rarer, and the left wing is squeezed.

This is the whole of the picture, and it is only directional. The naive estimate $R \sim 2^{\bar{k}}$ rises far more slowly than the data (Fig. 1, right): $2^{\bar{k}-1}$ runs from 4 to 9.4 while R runs from 5 to 89. The shortfall is structural—the right wing is not merely “not-all-left” but also harvests the entire all-right family and every even-length mixed product—so R outruns $2^{\bar{k}}$. We name the qualitative picture a *topological entropy increase*: a low-entropy, fully ordered (single-colour) basin being absorbed into a high-entropy, mixed basin. We use the phrase only as a mnemonic for the measured monochromatic suppression; it names a counting asymmetry, not a dynamical law, and no entropy functional is claimed.

5 Honest ledger: the divergence rate is unidentifiable

The divergence of $R(X)$ is firm, and its sign and mechanism are explained. Its *rate* is not recoverable, and we state this without hedging.

Fitting the seven points to three candidate laws gives nearly indistinguishable residuals in $\log R$:

$$R \sim (\log X)^{2.9} \text{ (resid 0.12),} \quad R \sim X^{0.19} \text{ (resid 0.16),} \quad R \sim e^{1.5\sqrt{\log X}} \text{ (resid 0.13).}$$

No reasonable amount of additional data breaks this degeneracy. Two obstructions compound. First, the left-wing counts are tiny—1, 1, 2, 4, 6, 12, 17 across the decades—so the ratios carry Poisson relative errors of order 10–100%; the early points are essentially single observations. Second, Carmichael statistics are deeply pre-asymptotic: the Erdős count carries $\log \log \log / \log \log$ corrections that do not stabilise until astronomically large X , and the factor-count $\bar{k}(X)$ that drives the mechanism creeps upward by less than half a unit per several decades. Extending the census by one or two further decades would roughly double the left-wing count, nowhere near enough to separate a power of $\log X$ from a small power of X . Accordingly we report $R(X)$ as a *measured divergent trend* with a structural cause, and we make *no* claim of an asymptotic divergence law: in any computational window a human can reach, that law is unidentifiable.

6 Conclusion

Read on the $6N$ skeleton, the Carmichael numbers exhibit a rigid one-way membrane (left-wing numbers are monochromatically left, Proposition 1) and a parity selection rule (Proposition 2), both verified to 10^{10} with zero violations. The geometric consequence is an asymmetric collapse of the left-wing population, with the right/left ratio rising from 5.0 to 88.8 across seven decades, driven by the exponential cost of the monochromatic constraint as the typical prime-factor count grows. We claim only this: the two structural laws are elementary Korselt consequences, cleanly stated in wing coordinates and confirmed empirically; the collapse is a genuine measured phenomenon with a structural mechanism; and its precise asymptotic rate is, by Poisson sparsity and pre-asymptotic drift, not identifiable. The framework says nothing about the infinitude or the true counting function of Carmichael numbers beyond what [3, 4, 5] already give.

Data and code availability. The complete enumeration to 10^{10} (the Korselt divisor method, the wing census, the membrane and parity verifiers, and the figure) is openly available at <https://github.com/Ruqing1963/6N-carmichael-collapse>.

References

- [1] A. Korselt, *Problème chinois*, L’intermédiaire des mathématiciens **6** (1899), 142–143.
- [2] R. D. Carmichael, *Note on a new number theory function*, Bull. Amer. Math. Soc. **16** (1910), 232–238.
- [3] W. R. Alford, A. Granville, C. Pomerance, *There are infinitely many Carmichael numbers*, Ann. of Math. **139** (1994), 703–722.
- [4] P. Erdős, *On pseudoprimes and Carmichael numbers*, Publ. Math. Debrecen **4** (1956), 201–206.
- [5] R. G. E. Pinch, *The Carmichael numbers up to 10^{21}* , Proc. Conf. Algorithmic Number Theory (2007); tables at <http://www.s369624816.websitehome.co.uk/rgep/cartable.html>.

- [6] R. Chen, *Arithmetic geodynamics on the $6N$ skeleton: a review of Parts I–XIX*, Zenodo, doi:10.5281/zenodo.20585301 (2026).
- [7] R. Chen, *Fine structure of the wing-transition memory on the $6N$ skeleton (Part XXXV)*, Zenodo, doi:10.5281/zenodo.20603759 (2026).